

Chemistry exam-21 — Marking Report

Chapters 1–9 · 46 marks, no optional questions · attempted 9 September 2026 · independently deep-checked

41

out of 46 · 89.1%

16/16

Sections B + C

83%

MCQs

The short version. Seemab scored 41 out of 46, her strongest paper since August. Her written work was **perfect** — both short questions and all six parts of the long numerical, sixteen marks out of sixteen. All five lost marks are multiple choice, and all five are specific facts rather than method failures.

Two results matter more than the score. First, the habit that has cost her marks on six consecutive papers — doing the mathematics and then not writing the concluding sentence — **cost her nothing here**. Every part of every question ends with an explicit, labelled final answer. Second, her multiple choice climbed from 60% on Monday's maths paper to 83% here, the first movement since that weakness was identified.

The five MCQs she missed

Q	Topic	Her answer	Correct	The error
9	Quantum mechanical model (Ch3)	Fixed circular orbits	Probability regions called orbitals	She gave the Bohr model. The quantum mechanical model replaced fixed orbits with regions of probability.
13	Electron affinity (Ch4)	Energy needed to remove the outermost electron	Energy released when a gaseous atom gains an electron	That is the definition of ionization energy . The two are opposites: one removes an electron, the other gains one.
14	Haber process catalyst (Ch4)	Nickel	Iron	Straight recall. Nickel is the catalyst for hydrogenating alkenes, not for ammonia.
19	Why diamond's melting point is so high (Ch5)	Small individual molecules held by weak forces	Giant covalent (network) structure	She described a simple molecular solid, which would melt easily. Diamond is one continuous covalent network, which is exactly why it does not.
27	How a catalyst works (Ch8)	Increases the overall enthalpy change	Provides a pathway with lower activation energy	A genuine misconception worth correcting directly: a catalyst never changes ΔH . It lowers the barrier, not the energy difference between reactants and products.

Questions 13 and 14 look transposed against the key, which is the signature of a mis-numbered answer list. It was formally tested: her answers score 25 matches at the correct alignment, against 2, 1, 2 and 0 at shifts of -2 , -1 , $+1$ and $+2$. There is no misalignment; these are two independent errors that happen to look like a swap.

Section B — 8/8

Q	Topic	Mark	Comment
2(i)	Burning hydrogen	4/4	$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$, correct coefficients and state symbols, classified exothermic, with the reason given in terms of bond breaking absorbing energy and bond forming releasing more. Never set in any of the 20 earlier chemistry papers.
2(ii)	Aluminium sulfate	4/4	LCM of 3 and 2 = 6, giving 2 and 3, ratio 2:3, criss-cross to $\text{Al}_2(\text{SO}_4)_3$ with correct bracket notation. Also never set before.

Section C — 8/8

A six-part numerical chain on an unknown dicarboxylic acid, every step correct:

Part	Required	Her answer	Mark
(a)	Empirical formula from 49.3% C, 6.8% H, 43.9% O	$\text{C}_3\text{H}_5\text{O}_2$	3/3
(b)	Molecular formula given $M_r = 146$	$\text{C}_6\text{H}_{10}\text{O}_4$	1/1
(c)	Moles in a 43.8 g sample	0.300 mol	1/1
(d)	Number of molecules	1.807×10^{23}	1/1
(e)	Balanced combustion equation	$2\text{C}_6\text{H}_{10}\text{O}_4 + 13\text{O}_2 \rightarrow 12\text{CO}_2 + 10\text{H}_2\text{O}$	1/1
(f)	Mass of CO_2 from 0.3 mol	79.2 g	1/1

What this paper shows

1. The missing-conclusion habit did not cost her anything. That streak ran six papers — Maths exam-10, exam-13, exam-15, exam-16 twice, exam-17 three times, Chemistry exam-19 and exam-20. Here every sub-part ends with an explicit, correctly labelled final answer: the classification and reason are stated, the bracketed formula is written out, and each of the six numerical parts closes with a value and its unit. This was the single highest-value change available to her and she has made it. **Tell her this.**

2. Her multiple choice moved for the first time. 60% on maths exam-17 on Monday, 83% here. One paper in a different subject is not proof, but the advice to write rough working rather than answer mentally went out with this paper, and the number went up. Worth reinforcing while the link is fresh.

3. All five losses are recall, not reasoning. None of them is a method failure or a calculation slip. Two are definitions confused with their opposite (electron affinity vs ionization energy; simple molecular vs giant covalent), one is a model she has mixed up with an older one, one is a named catalyst, and one is a genuine misconception about what a catalyst does to ΔH . These are learnable in an evening with flashcards.

One thing to mention gently

In part (c) her written working shows the numerator as **43.6**, while the question gives 43.8 g. Her final answer of 0.300 mol is correct and only follows from 43.8, so she used the right number and mis-copied it onto the page. No mark was affected. It is worth a word because a mis-copied figure usually does not stay harmless — here it happened not to propagate.

Similarly, in a charge check she volunteered on the aluminium sulfate question she wrote $2(+3) + 2(-2)$ where the second coefficient should be 3, and still concluded $6 + (-6) = 0$. No marks were deducted, since the question asked only for the LCM step, the formula and the brackets, and all three were already correct and complete before this bonus line. But she is not yet reliably separating a formula coefficient from the ion's own charge when checking under time pressure.

Revision priorities

- **A catalyst does not change ΔH .** It provides a lower-activation-energy pathway. This is the one item on the paper that is a true misconception rather than a forgotten fact.
- **Electron affinity versus ionization energy.** Learn them as an opposing pair, not separately: one is energy released on gaining an electron, the other energy needed to remove one.
- **Bonding and structure determine melting point.** Giant covalent networks (diamond, silicon dioxide) melt very high; simple molecular solids melt low.
- **The quantum mechanical model versus Bohr's.** Orbits were replaced by orbitals, which are probability regions.
- **Iron is the Haber process catalyst.** Straight memorisation.

Marking was independently deep-checked. All 30 MCQ answers were re-derived from the printed question stems rather than the answer key, and no key error was found. Her answer list was re-read from the scan letter by letter, and a shift guard was run across five alignments. Both passes reached 41/46 with no disagreements.